

CAPS INSPECTION

Cap Cavity Number Preprocessing

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Project Work for Computer Vision and Image Processing M

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Guide to project files

Our project is structured in multiple files, each one has quite a similar content but is thought to give insights over different aspects of the project.

This is the file list:

- **cnp.ipynb** This is the main file, multiple alternatives to the solution of each problem that we faced are presented and the effects on the whole dataset are shown each time. Furthermore test on corner cases are performed to test the robustness.
- **single_cap.ipynb** This file is meant to show step by step the chosen strategies and their effects on a single cap, to better allow to visualize the flow of the process of the chosen solution.
- **cap_preprocessing.py** This file is meant to provide a sort of API, in which the whole process or the single preprocessing steps can be selectively called.

Guide to project datasets

Our project repository includes two datasets:

- The one contained in `./img/` which is the given dataset.
- The one contained in `./custom_imgs/`, which is the one we created to test the robustness of our algorithm w.r.t. some corner cases.